



2011
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

Extension 1

General Instructions

- Reading Time- 5 minutes
- Working Time – 2 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question in a new booklet.

Total marks (84)

- Attempt Questions 1-7
- All questions are of equal value

Total Marks – 84
Attempt Questions 1-7
All Questions are of equal value

QUESTION 1	(12 MARKS)	Begin a NEW booklet.	Marks
a)	Calculate $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$.		1
b)	Solve $\frac{x}{2x-1} \geq 3$.		3
c)	If α, β, γ are the roots of the equation $4x^3 - 6x^2 - 3x + 8 = 0$, find the value of		
	(i) $\alpha\beta + \alpha\gamma + \beta\gamma$		1
	(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.		2
d)	If $\log_5 10 = 2.48$ find the exact value of $\log_5 4$.		2
e)	Use the substitution $u = x^2 - 3$ to evaluate $\int_2^6 \frac{x}{\sqrt{x^2 - 3}} dx$.		3

QUESTION 2 (12 MARKS) Begin a NEW booklet.

a) (i) Prove that $\frac{1 + \cos 2\theta}{\sin 2\theta} = \cot \theta$. 2

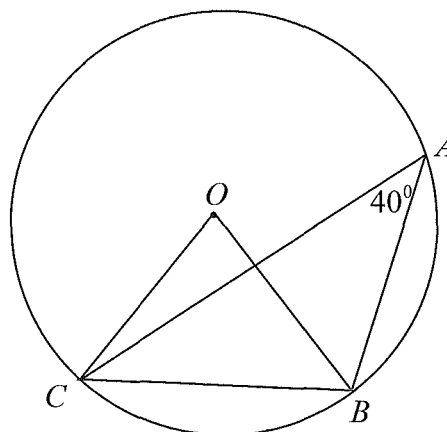
(ii) Hence calculate the exact value of $\cot \frac{\pi}{12}$. 2

b) A polynomial is given by $P(x) = x^3 + ax^2 + bx + 8$. 3

Determine the values of a and b if $(x + 4)$ is a factor of $P(x)$
and 18 is the remainder when $P(x)$ is divided by $(x + 1)$.

c) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{6}} \frac{dx}{6 + 3x^2}$. 3

d) In the diagram below A , B and C are points on the circumference of a circle centre O . If $\angle CAB = 40^\circ$, find the size of $\angle OBC$ giving reasons for your answer. 2



QUESTION 3 (12 MARKS) Begin a NEW booklet.

Marks

- a) Use one application of Newton's method to find a second approximation to the root of the equation $3 \sin x - 2x = 0$, by taking 1.56 as your first approximation.

3

Write your answer correct to 2 decimal places.

- b) Find the term independent of x in the expansion of

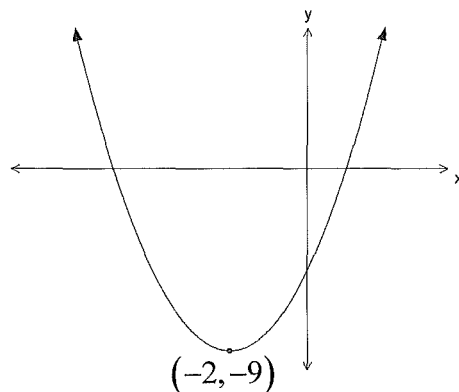
3

$$\left(x^2 - \frac{1}{x^3}\right)^{10}.$$

- c) Show that the function $f(x) = \frac{e^x}{4 - e^x}$ is monotonically increasing over the domain of x .

2

- d) The graph of $g(x) = x^2 + 4x - 5$ is shown in the diagram.



- (i) Sketch the graph of the inverse function of $g(x) = x^2 + 4x - 5$, for $x \geq -2$.

1

- (ii) State the domain of the inverse function $g^{-1}(x)$.

1

- (iii) Find an expression for $y = g^{-1}(x)$ in terms of x .

2

QUESTION 4 (12 MARKS) Begin a NEW booklet.

Marks

- a) Find $\int \sin \theta \cos^2 \theta \, d\theta$ by using the substitution $u = \cos \theta$.

2

- b) Evaluate $\sin \left[\tan^{-1}(-\sqrt{3}) \right]$.

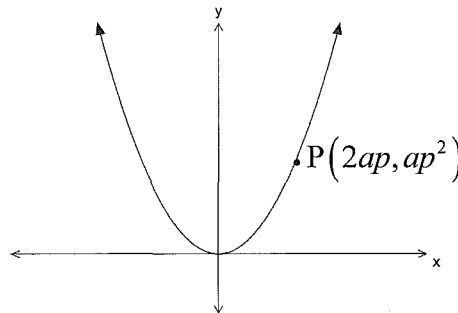
2

- c) A spherical balloon is inflated at a constant rate of $12 \cdot 6 \text{ cm}^3 / \text{s}$.
At what rate is the surface area increasing when the radius of the balloon is 12 cm?

3

$$SA = 4\pi r^2 \text{ and } V = \frac{4}{3}\pi r^3.$$

- d) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$ as shown in the diagram drawn below.



The equation of the normal to the curve at P is $x + py = 2ap + ap^3$.
DO NOT prove this.

- (i) Find the co-ordinates of the point Q where the normal at P meets the y-axis.

1

- (ii) Show that the co-ordinates of the point R, which divides the interval PQ externally in the ratio 1 : 2 are given by $(4ap, ap^2 - 2a)$

2

- (iii) Find the Cartesian equation of the locus of R.

2

QUESTION 5 (12 MARKS) Begin a NEW booklet.

Marks

a) Consider the function $y = 4 \sin^{-1} \left(\frac{x}{3} \right)$

(i) State the domain and range of the function.

2

(ii) Sketch the graph of the function showing all essential features.

1

(iii) Calculate the gradient of the tangent to the curve at the point where $x = \sqrt{5}$.

2

b) The area bounded by the curve $y = \sin 2x$,
the x -axis and the line $x = \frac{\pi}{4}$ is rotated about the x -axis.

3

Calculate the exact volume of the solid of revolution.

c) The rate of growth of bacteria in a culture is given by $\frac{dN}{dt} = k(N - 800)$,
where N is the number of bacteria and t is time, in seconds.

(i) Show that $N = 800 + Ae^{kt}$ is a solution of this equation.

1

(ii) Initially there are 1 000 bacteria and five seconds later there are 1 700 bacteria present in the culture. Calculate the number of bacteria present after ten seconds.

3

QUESTION 6 (12 MARKS) Begin a NEW booklet.

Marks

- a) (i) Express $\sqrt{3} \sin \theta - \cos \theta$ in the form $R \sin(\theta - \alpha)$

2

where R is positive and α is acute.

- (ii) Hence solve $\sqrt{3} \sin \theta - \cos \theta = -1$ for $0 \leq \theta \leq 2\pi$.

2

- b) Write the binomial expansion of $(3a - 2b)^4$ in simplified form.

2

- c) Use the table of standard integrals to show that $\int_6^{10} \frac{dx}{\sqrt{x^2 - 36}} = \log_e 3$.

2

- d) Use the principle of Mathematical Induction to prove that for all positive integers

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

4

QUESTION 7 (12 MARKS) Begin a NEW booklet.

Marks

- a) The acceleration of a particle P is given by $\ddot{x} = 4x(x^2 - 1)$,

3

where x is the displacement of the particle from the origin, in metres, after t seconds. Initially the particle is at the origin, moving to the right with a velocity of $\sqrt{2}$ m/s.

Prove that the velocity of the particle is $v = -\sqrt{2}(x^2 - 1)$.

- b) Consider the expansion of $\left(x + \frac{3}{x^2}\right)^8$ with the general term T_{k+1} .

(i) Show that $\frac{T_{k+1}}{T_k} = \frac{9-k}{k} \times \frac{3}{x^3}$

3

- (ii) Hence calculate the greatest co-efficient in the expansion.

2

- c) A particle is moving in simple harmonic motion about a fixed point, with a velocity measured in metres/second, given by $v^2 = 21 + 4x - x^2$.

- (i) Between what two points is the particle oscillating?

1

- (ii) What is the centre of the motion?

1

- (iii) Write the amplitude of the motion.

1

- (iv) Calculate the particle's maximum speed.

1

QUESTION 1.

$$\begin{aligned}
 \text{a) } \lim_{x \rightarrow 0} \frac{\tan 3x}{2x} &= \frac{3}{2} \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \\
 &= \frac{3}{2} \times 1 \\
 &= \frac{3}{2}
 \end{aligned}$$

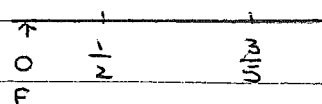
$$\text{b) } \frac{x}{2x-1} \geq 3, \quad x \neq \frac{1}{2}$$

$$x(2x-1) \geq 3(2x-1)^2$$

$$3(2x-1)^2 - x(2x-1) \leq 0$$

$$(2x-1)(6x-3-x) \leq 0$$

$$(2x-1)(5x-3) \leq 0$$



$$\therefore \frac{1}{2} < x \leq \frac{3}{5}$$

$$\begin{aligned}
 \text{c) (i) } \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\
 &= -\frac{3}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} & \alpha\beta\gamma &= \frac{-d}{a} \\
 & & &= -2 \\
 &= -\frac{3}{4} \div -2 \\
 &= \frac{3}{8}
 \end{aligned}$$

$$\text{d) } \log_5 10 = 2.48$$

$$\log_5 (2 \times 5) = 2.48$$

$$\log_5 2 + \log_5 5 = 2.48$$

$$\log_5 2 = 1.48$$

$$2 \log_5 2 = 2.96$$

$$\therefore \log_5 4 = 2.96$$

$$e) \int_2^6 \frac{x}{\sqrt{x^2-3}} dx$$

$$u = x^2 - 3$$

$$du = 2x dx$$

$$= \frac{1}{2} \int_1^{33} u^{-1/2} du$$

$$x=2, \quad u=1$$

$$x=6, \quad u=33$$

$$= \frac{1}{2} \cdot 2 \left[u^{\frac{1}{2}} \right]_1^{33}$$

$$= \sqrt{33} - 1$$

QUESTION 2

$$a) \quad (i) \quad LHS = \frac{1 + \cos 2\theta}{\sin 2\theta}$$

$$= \frac{1 + 2\cos^2\theta - 1}{2\sin\theta\cos\theta}$$

$$= \frac{2\cos^2\theta}{2\sin\theta\cos\theta}$$

$$= \frac{\cos\theta}{\sin\theta}$$

$$= \cot\theta$$

$$= RHS$$

$$(ii) \quad \cot\theta = (1 + \cos 2\theta) \div \sin 2\theta$$

$$\cot \frac{\pi}{12} = \left(1 + \cos \frac{\pi}{6}\right) \div \sin \frac{\pi}{6}$$

$$= \left(1 + \frac{\sqrt{3}}{2}\right) \div \frac{1}{2}$$

$$= 2 + \sqrt{3}$$

$$b) \quad P(-4) = 0 \quad \text{ie} \quad -64 + 16a - 4b + 8 = 0$$

$$16a - 4b = 56$$

$$4a - b = 14 \quad \text{--- (1)}$$

$$P(-1) = 18 \quad \text{ie} \quad -1 + a - b + 8 = 18$$

$$a - b = 11 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2}$$

$$3a = 3$$

$$\therefore a = 1, \quad b = -10$$

$$\begin{aligned}
 \text{c) } \int_{\sqrt{2}}^{\sqrt{6}} \frac{dx}{6+3x^2} &= \frac{1}{3} \int_{\sqrt{2}}^{\sqrt{6}} \frac{dx}{2+x^2} & a^2 &= 2 \\
 & & a &= \sqrt{2} \\
 &= \frac{1}{3} \cdot \frac{1}{\sqrt{2}} \left[\tan^{-1} \frac{x}{\sqrt{2}} \right]_{\sqrt{2}}^{\sqrt{6}} \\
 &= \frac{1}{3\sqrt{2}} \left(\tan^{-1} \sqrt{3} - \tan^{-1} 1 \right) \\
 &= \frac{1}{3\sqrt{2}} \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\
 &= \frac{\pi}{36\sqrt{2}}
 \end{aligned}$$

d) $\angle COB = 80^\circ$ (angle at the centre is twice the angle at the circumference, subtended by the same arc)

$\triangle OCB$ is isosceles as $OC = OB$ (radii)

$\therefore \angle OBC = 50^\circ$ (base angle in isosceles triangle)

QUESTION 3

a) Let $f(x) = 3 \sin x - 2x$ $x_1 = 1.56$

$$f'(x) = 3 \cos x - 2$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.56 - \left(\frac{3 \sin 1.56 - 2 \times 1.56}{3 \cos 1.56 - 2} \right)$$

$$\approx 1.4989 \dots$$

$\therefore$ a second approxⁿ to the root is 1.50, to 2 dec. pl.

b) $T_{k+1} = {}^nC_k a^{n-k} \cdot b^k$

$$= {}^{10}C_k (x^2)^{10-k} \cdot (-1)^k \cdot (x^{-3})^k$$

$$20 - 2k - 3k = 0$$

$$k = 4$$

$$T_{k+1} = {}^{10}C_4 (-1)^4$$

$\therefore$ the independent term is 210

c) If $f(x)$ is monotonically increasing for all x , then $f'(x) > 0$ for all values of x

$$f(x) = \frac{e^x}{4 - e^x}$$

$$u = e^x$$

$$v = 4 - e^x$$

$$u' = e^x$$

$$v' = -e^x$$

$$f'(x) = \frac{(4 - e^x)^x e^x + e^x \cdot e^x}{(4 - e^x)^2}$$

$$= \frac{4e^x}{(4 - e^x)^2}$$

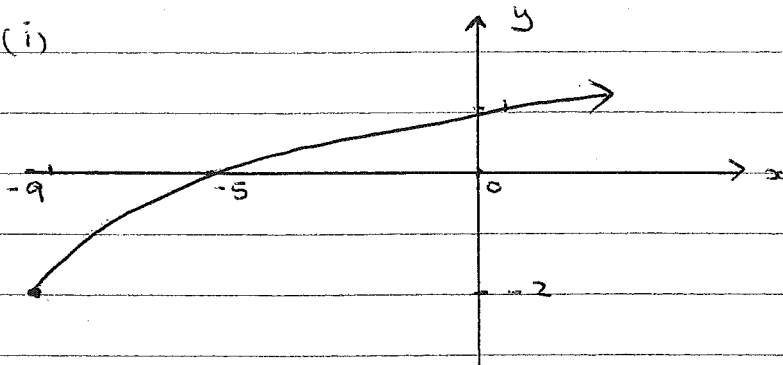
CONT.

Since $e^x > 0$ for all x
and $(4 - e^x)^2 > 0$ for all x

then $f'(x) > 0$

and the function is increasing over the domain of x

d) (i)



(ii) $x \geq -9$

(iii) $x = y^2 + 4y - 5$

$$x + 5 = y^2 + 4y$$

$$x + 5 + 4 = y^2 + 4y + 4$$

$$x + 9 = (y + 2)^2$$

$$y + 2 = \pm \sqrt{x + 9}$$

$\therefore y = -2 + \sqrt{x + 9}$ is the inverse fn

QUESTION 4

a) $\int \sin \theta \cos^2 \theta \, d\theta$

$$u = \cos \theta$$

$$du = -\sin \theta \, d\theta$$

$$= - \int u^2 \, du$$

$$= -\frac{1}{3} u^3 + c$$

$$= -\frac{1}{3} \cos^3 \theta + c$$

b) $\sin \left[\tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) \right] = \sin \left(-\frac{\pi}{3} \right)$

$$= -\frac{\sqrt{3}}{2}$$

c) $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$

$$V = \frac{4}{3} \pi r^3$$

$$12.6 = 4\pi \times 12^2 \times \frac{dr}{dt}$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dr}{dt} = \frac{12.6}{4\pi \times 12^2}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$A = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$= 8\pi \times 12 \times \frac{12.6}{4\pi \times 12^2}$$

$\therefore$ surface area is increasing at the rate of $2.1 \text{ cm}^2/\text{s}$

d) over page

d) (i) when $x = 0$, $py = 2ap + ap^3$
 $y = 2a + ap^2$

$\therefore Q$ is the point $(0, 2a + ap^2)$

(ii) $P(\overset{x_1}{2ap}, \overset{y_1}{ap^2})$ $Q(\overset{x_2}{0}, \overset{y_2}{2a + ap^2})$ $\overset{m}{-1} : \overset{n}{2}$

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right) = \left(\frac{0 + 4ap}{1}, \frac{-2a - ap^2 + 2ap^2}{1} \right)$$

$\therefore R$ is the point $(4ap, ap^2 - 2a)$

(iii) $x = 4ap$ — (1)

$y = ap^2 - 2a$ — (2)

from (1) $p = \frac{x}{4a}$

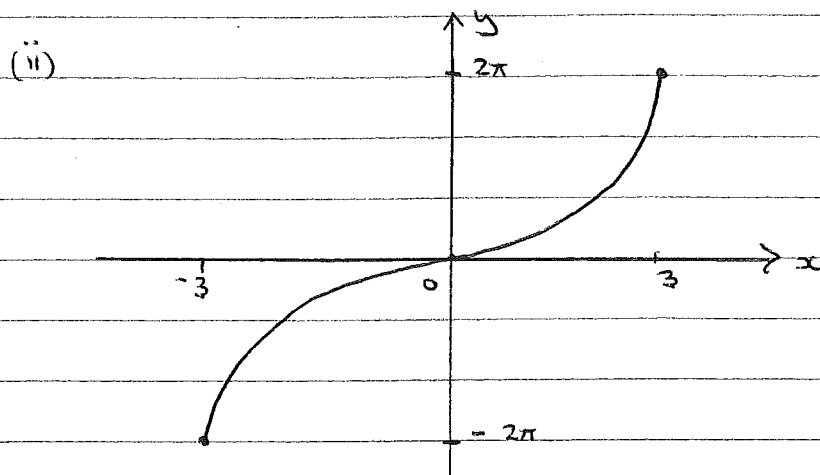
sub in (2) $y = \frac{a \cdot x^2}{16a^2} - 2a$

$$y + 2a = \frac{x^2}{16a}$$

$\therefore x^2 = 16a(y + 2a)$ is the eqn of the locus of R

QUESTION 5

- a) (i) domain is $-3 \leq x \leq 3$
range is $-2\pi \leq y \leq 2\pi$



(iii) $y = 4 \sin^{-1}\left(\frac{x}{3}\right)$

$$\frac{dy}{dx} = 4 \cdot \frac{1}{\sqrt{9-x^2}}$$

when $x = \sqrt{5}$, $\frac{dy}{dx} = 4 \cdot \frac{1}{\sqrt{9-5}}$

$\therefore$ gradient of the tangent is 2

b) $V = \pi \int_{\pi/4}^{\pi/4} \sin^2 2x \, dx$

$$= \frac{\pi}{2} \int_0^{\pi/4} (1 - \cos 4x) \, dx$$

$$= \frac{\pi}{2} \left[x - \frac{1}{4} \sin 4x \right]_0^{\pi/4}$$

$$= \frac{\pi}{2} \left[\frac{\pi}{4} - \frac{1}{4} \sin \pi - (0 - 0) \right]$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\sin^2 2x = \frac{1}{2}(1 - \cos 4x)$$

$\therefore$ the volume is $\frac{\pi^2}{8}$ cubic units

$$c) \quad (i) \quad N = 800 + A e^{kt} \quad \text{ie} \quad A e^{kt} = N - 800$$

$$\frac{dN}{dt} = k \cdot A e^{kt}$$

$$\text{ie/} \quad \frac{dN}{dt} = k(N - 800)$$

$$(ii) \quad N = 800 + A e^{kt}$$

$$t = 0, N = 1000 \quad \text{so} \quad 1000 = 800 + A$$

$$A = 200$$

$$N = 800 + 200 e^{kt}$$

$$t = 5, N = 1700$$

$$1700 = 800 + 200 e^{5k}$$

$$\frac{900}{200} = e^{5k}$$

$$5k = \log_e 4.5$$

$$k = \frac{1}{5} \log_e 4.5$$

$$\text{when } t = 10$$

$$N = 800 + 200 e^{2 \log_e 4.5}$$

$$= 800 + 200 (4.5)^2$$

$$= 4850$$

$\therefore$ there are 4850 bacteria after 10s.

QUESTION 6

$$a) \quad (i) \quad \sqrt{3} \sin \theta - \cos \theta = R \sin \theta \cos \alpha - R \cos \theta \sin \alpha$$

$$R \cos \alpha = \sqrt{3} \quad \text{--- (1)}$$

$$R \sin \alpha = 1 \quad \text{--- (2)}$$

$$\textcircled{2} \div \textcircled{1} \quad \tan \alpha = 1/\sqrt{3}$$

$$\alpha = \pi/6$$

$$\textcircled{1}^2 + \textcircled{2}^2 \quad R^2 = 4$$

$$R = 2$$

$$\therefore \sqrt{3} \sin \theta - \cos \theta = 2 \sin(\theta - \frac{\pi}{6})$$

$$(ii) \quad 2 \sin(\theta - \frac{\pi}{6}) = -1 \quad \text{for} \quad -\frac{\pi}{6} \leq \theta - \frac{\pi}{6} \leq 2\pi - \frac{\pi}{6}$$

$$\sin(\theta - \frac{\pi}{6}) = -\frac{1}{2}$$

S	A
T	C
✓	✓

$$\theta - \frac{\pi}{6} = -\frac{\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore \theta = 0, \frac{4\pi}{3}, 2\pi$$

$$b) \quad (3a - 2b)^4$$

$$= (3a)^4 + 4(3a)^3(-2b) + 6(3a)^2(-2b)^2 + 4(3a)(-2b)^3 + (-2b)^4$$

$$= 81a^4 - 216a^3b + 216a^2b^2 - 96ab^3 + 16b^4$$

$$c) \quad \int_6^{10} \frac{dx}{\sqrt{x^2 - 36}} = \left[\ln(x + \sqrt{x^2 - 36}) \right]_6^{10}$$

$$= \ln(10 + \sqrt{100 - 36}) - \ln 6$$

$$= \ln 18 - \ln 6$$

$$= \ln(18/6)$$

$$= \ln 3$$

d) Step 1 Prove true when $n = 1$

$$\begin{aligned} \text{LHS} &= \frac{1}{1 \times 3} \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \frac{1}{2+1} \\ &= \frac{1}{3} \end{aligned}$$

$\therefore$ result is true when $n = 1$

Step 2 Assume true when $n = k$

$$\text{i.e. } S_k = \frac{k}{2k+1}$$

Prove true when $n = k+1$

$$\text{i.e. } S_{k+1} = \frac{k+1}{2k+3}$$

Proof: $S_{k+1} = S_k + T_{k+1}$

$$= \frac{k}{(2k+1)} + \frac{1}{(2k+1)(2k+3)}$$

$$= \frac{k(2k+3) + 1}{(2k+1)(2k+3)}$$

$$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$$

$$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$$

$$\therefore S_{k+1} = \frac{k+1}{2k+3}$$

Step 3 Since the result is true when $n = 1$, it is also true when $n = 1+1$ i.e. $n = 2$

Since result is true when $n = 2$, it is also true when $n = 2+1$ i.e. $n = 3$ etc

$\therefore$ result is true for all positive integers.

QUESTION 7

a)

$$\ddot{x} = 4x(x^2 - 1)$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4x^3 - 4x$$

$$\frac{1}{2} v^2 = x^4 - 2x^2 + c$$

$$x=0, v=\sqrt{2}$$

$$\frac{1}{2} \cdot 2 = 4 - 2 \cdot 2 + c \Rightarrow c=1$$

$$\frac{1}{2} v^2 = x^4 - 2x^2 + 1$$

$$\frac{1}{2} v^2 = (x^2 - 1)^2$$

$$v^2 = 2(x^2 - 1)^2$$

$$v = \pm \sqrt{2} (x^2 - 1)$$

$$\text{when } x=0, v=\sqrt{2}$$

$$\therefore v = -\sqrt{2} (x^2 - 1)$$

$$\begin{aligned} \text{b) (i) } \frac{T_{k+1}}{T_k} &= \binom{8}{k} x^{8-k} \cdot 3^k \cdot x^{-2k} \div \left[\binom{8}{k-1} x^{9-k} \cdot 3^{k-1} \cdot x^{-2k+2} \right] \\ &= \frac{8!}{k! (8-k)!} \times \frac{(k-1)! (9-k)!}{8!} \times 3 \times x^{-3} \\ &= \frac{9-k}{k} \cdot \frac{3}{x^3} \end{aligned}$$

$$\text{(ii) } \frac{T_{k+1}}{T_k} > 1$$

$$\frac{3(9-k)}{k} > 1$$

$$27 - 3k > k$$

$$4k < 27$$

$$k < 6 \frac{3}{4}$$

$$\therefore k = 6$$

$$\text{co-efficient} = \binom{8}{6} \cdot 3^6$$

$\therefore$ greatest co-efficient is 20412

c) (i) $v = 0$ at the end points of the motion

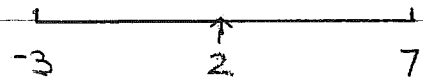
$$x^2 - 4x - 21 = 0$$

$$(x - 7)(x + 3) = 0$$

$$x = 7, -3$$

$\therefore$ particle is oscillating between $x = -3$ and $x = 7$

(ii) Centre is at $x = 2$



(iii) Amplitude is 5 m

(iv) Max speed occurs at the centre of the motion

$$\text{when } x = 2, \quad v^2 = 21 + 8 - 4$$

$$v^2 = 25$$

$\therefore$ max speed is 5 m/s